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Metal-insulator-metal capacitor

Abstract

A metal-insulator-metal capacitor includes a first electrode disposed in a first region of an upper surface of a substrate, a second electrode covering the first electrode and extending to a second region surrounding an outer periphery of the first region, a third electrode covering the second electrode and extending to a third region surrounding an outer periphery of the second region, a first dielectric layer disposed between the first electrode and the second electrode to cover an upper surface and a side surface of the first electrode and extending to the second region, and a second dielectric layer disposed between the second electrode and the third electrode to cover an upper surface and a side surface of the second electrode and extending to the third region and in contact with the first dielectric layer.

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Background/Summary

CROSS TO REFERENCE TO RELATED APPLICATION(S) (1) This is a divisional of U.S. non-provisional patent application Ser. No. 17/599,176, filed Dec. 22, 2021, which claims benefit of priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0066797, filed on May 25, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD
(1) Example embodiments of the present disclosure relate to a metal-insulator-metal capacitor.
BACKGROUND
(2) Semiconductor devices based on Bipolar, BiCMOS and CMOS techniques may require integrated capacitors having high voltage linearity, and low parasitic capacitance. However, a general MOS capacitor may have issues in that voltage linearity may be low due to a space charge region for voltage induction, and also, a large amount of parasitic capacitance may be present therein.
(3) To address the issues above, a metal-insulator-metal capacitor may be used. Such a metal-insulator-metal capacitor may mainly be used to store electrical charges in various semiconductor devices such as a mixed-signal product and an analog product.
SUMMARY
(4) Some example embodiments of the present disclosure provide metal-insulator-metal capacitors having improved reliability and yield.
(5) In an example embodiment of the present disclosure, a metal-insulator-metal capacitor includes a first electrode, a second electrode, a third electrode, a first dielectric layer, and a second dielectric layer. The first electrode is in contact with an upper surface of a substrate and disposed in a first region of the upper surface. The second electrode covers the first electrode and extends to a second region surrounding an outer periphery of the first region. The third electrode covers the second electrode and extends to a third region surrounding an outer periphery of the second region. T first dielectric layer is disposed between the first electrode and the second electrode to cover an upper surface and a side surface of the first electrode and extends to the second region of the upper surface of the substrate. The second dielectric layer is disposed between the second electrode and the third electrode to cover an upper surface and a side surface of the second electrode and extends to the third region of the upper surface of the substrate and is in contact with the first dielectric layer.
(6) In an example embodiment of the present disclosure, a metal-insulator-metal capacitor includes first to nth electrodes stacked on a substrate. The nth electrode covers an upper surface and a side surface of the n−1 electrode stacked therebelow. The metal-insulator-metal capacitor also includes first to n−1th dielectric layers interposed between the first to nth electrodes, and covering upper surfaces and side surfaces of first to n−1th electrodes, respectively. The first dielectric layer is in contact with the substrate. Second to n−1 dielectric layers have regions in contact with first to

n−2th dielectric layers stacked therebelow, respectively. The nth electrode has first to n−1th bent portions formed in a staircase shape in an edge region thereof.
(7) In an example embodiment of the present disclosure, a metal-insulator-metal capacitor includes a first electrode disposed on an upper surface of a substrate, a second electrode covering an upper surface and a side surface of the first electrode, a third electrode covering a side surface and an upper surface of the second electrode, a first dielectric layer interposed between the first electrode and the second electrode, and a second dielectric layer interposed between the second electrode and the third electrode. An interior angle between the upper surface and the side surface of each of the first electrode, the second electrode and the third electrode is in a range of 90° to 135°.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a cross-sectional diagram illustrating a metal-insulator-metal capacitor according to an example embodiment of the present disclosure;
(3) FIG. **2** is a plan diagram illustrating a metal-insulator-metal capacitor according to an example embodiment of the present disclosure;
(4) FIG. **3** is a cross-sectional diagram illustrating a metal-insulator-metal capacitor according to an example embodiment of the present disclosure;
(5) FIG. **4** is a cross-sectional diagram illustrating a metal-insulator-metal capacitor according to an example embodiment of the present disclosure; and
(6) FIG. **5**, FIG. **6**, FIG. **7**, FIG. **8**, FIG. **9**, FIG. **10**, FIG. **11** and FIG. **12** are each cross-sectional diagrams illustrating a method of manufacturing the metal-insulator-metal capacitor illustrated in FIG. **1**.
DETAILED DESCRIPTION
(7) Hereinafter, example embodiments of the present disclosure will be described as follows with reference to the accompanying drawings.
(8) Terminology used herein to describe relative locational relationships between elements shown in the drawings should be considered in the context of what is shown in the drawings. For example, a description of a first element (e.g., an electrode or a dielectric layer) covering a second element (e.g., an electrode or a dielectric layer) may be taken to mean that the first element substantially or entirely overlaps at least one surface (e.g., an upper surface and perhaps one or more side surfaces) of the second element. A description of a first element (e.g., an electrode or a dielectric layer) overlapping a second element (e.g., an electrode or a dielectric layer) may be taken to mean that a view of the second element would be obstructed by the first element from one or more viewpoints. Descriptions of a first element being on a second element may be taken to mean that the first element is over the second element when an apparatus or device that includes the first element and the second element is arranged in practice with the same orientation as shown in the referenced drawing.
(9) Referring to FIG. **1** and FIG. **2**, a metal-insulator-metal capacitor **100** according to an example embodiment will be described. FIG. **1** is a cross-sectional diagram illustrating a metal-insulator-metal capacitor **100** according to an example embodiment. In FIG. **1**, the metal-insulator-metal capacitor **100** is included in a semiconductor device **1**. FIG. **2** is a plan diagram illustrating a metal-insulator-metal capacitor according to an example embodiment.
(10) In the semiconductor device **1**, a metal-insulator-metal capacitor **100** may be disposed on a substrate **10**. A cover layer **160** is disposed to covers the metal-insulator-metal capacitor **100**. The

first electrode **110**, the second electrode **130**, and the third electrode **150** of the metal-insulator-metal capacitor **100** may be electrically connected by the first via electrode **21** and the second via electrode **22**. A passivation layer **30** and a protective layer **40** may be stacked on the first via electrode **21** and the second via electrode **22**. A hole **50** penetrates through the passivation layer **30** and the protective layer **40** and is connected to the first via electrode **21** and the second via electrode **22**. The first via electrode **21** and the second via electrode **22** may be connected to the substrate **10** through the via hole V penetrating through the cover layer **160**, the first electrode **110**, the second electrode **130**, and the third electrode **150**, and the first dielectric layer **120** and the second dielectric layer **140**. The first via electrode **21** and the second via electrode **22** may be electrically connected to the first electrode **110**, the second electrode **130**, and the third electrode **150**. In an example embodiment, the first via electrode **21** is electrically connected to the first electrode **110** and the third electrode **150**, and the second via electrode **22** is electrically connected to the second electrode **130**. Also, in an example embodiment, the first electrode **110**, the second electrode **130**, and the third electrode **150** may be disposed. Example embodiments are not limited to the details described herein. For example, four or more electrodes may be disposed in an example embodiment.

(11) In the substrate **10**, a wiring region **12**, a first insulating layer **13**, and a second insulating layer **14** may be disposed on the semiconductor substrate **11**.

(12) The semiconductor substrate **11** may include, for example, a semiconductor element such as silicon or germanium (Ge), or a compound semiconductor such as silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), or indium phosphide (InP). The semiconductor substrate **11** may have a silicon on insulator (SOI) structure. A wiring region **12** for forming wiring may be disposed on the semiconductor substrate **11**. A wiring including a conductive layer **16** and a barrier layer **15** may be disposed in the wiring region **12**. The conductive layer **16** may be formed of a composite such as Ti/TiN/Al—Cu/Ti/TiN.

(13) A first insulating layer **13** and a second insulating layer **14** may be disposed on the semiconductor substrate **11**. The first insulating layer **13** and the second insulating layer **14** may be formed of different insulating materials. The first insulating layer **13** may include silicon nitride (SiN) and silicon oxynitride (SiON). The second insulating layer **14** may be formed of silicon oxide (SiO.sub.2).

(14) The metal-insulator-metal capacitor **100** may be disposed on the substrate **10**, and may include the first electrode **110**, the second electrode **130**, and the third electrode **150** and the first dielectric layer **120** and the second dielectric layer **140**.

(15) The first electrode **110**, the second electrode **130**, and the third electrode **150** may be stacked in order on the substrate **10**. The first electrode **110**, the second electrode **130**, and the third electrode **150** may be formed of the same conductive material, but may be formed of different conductive materials in example embodiments. The conductive material may include a metal such as aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt) and gold (Au), a metal silicide, a metal nitride, a metal oxide, polysilicon, conductive carbon, or a combination thereof.

(16) Referring to FIG. **2**, the first electrode **110** may be disposed in the first region A**1** of the upper surface **10**U of the substrate **10**. The first electrode **110** may be formed such that a side surface **110**S and an upper surface **110**U may have an interior angle (hereinafter, defined as a “first interior angle θ**1**”) of about 90° to about 135°.

(17) The second electrode **130** may be disposed to overlap the first electrode **110** in the first region A**1**, and may extend to the second region A**2** disposed around the first region A**1** and may be in contact with the upper surface **10**U of the substrate **10**. Accordingly, the second electrode **130** may entirely cover the upper surface **110**U and the side surface **110**S of the first electrode **110**. At least one instance of a first groove portion **131** may be formed in the second electrode **130** in the first region A**1**. The first via electrode **21** may be spaced apart from the internal side wall of the first groove portion **131**, may penetrate the first groove portion **131**, and may be in contact with the first

electrode **110** and the second electrode **130**. Accordingly, the first via electrode **21** may be insulated from the second electrode **130** and may be connected to the first electrode **110** and the third electrode **150**. The second electrode **130** may be stacked on the first electrode **110**, and may have a bent region bent in a staircase shape in a region in contact with the side surface **110**S of the first electrode **110**. The interior angle of the bent region (hereinafter, defined as a “second interior angle θ**2**”) may be the same as the first interior angle θ**1**, or may be greater by about 10° degrees or less. In other words, an angle difference θ**2**−θ**1** between the second interior angle θ**2** and the first interior angle θ**1** may be in a range from about 0° to about 10°.

(18) The third electrode **150** may be disposed to overlap the second electrode **130** in the first region A**1** and the second region A**2**, may extend to the third region A**3** disposed around the second region A**2**, and may overlap the substrate **10**. Accordingly, the second electrode **130** may entirely cover the first electrode **110**, and the third electrode **150** may entirely cover the second electrode **130**. In the third electrode **150**, at least one instance of a second groove portion **151** may be formed in the second region A**2**. The second via electrode **22** may be spaced apart from the internal side wall of the second groove portion **151**, may penetrate the second groove portion **151**, and may be in contact with the second electrode **130**. The third electrode **150** may be stacked on the second electrode **130**, and may have a bent region bent in a staircase shape in a region in contact with the side surface **130**S of the second electrode **130**. The interior angle of the bent region (hereinafter, defined as a “third interior angle θ**3**”) may be the same as the second interior angle θ**2**, or may be greater by about 10° degrees or less. In other words, an angle difference θ**3**−θ**2** between the third interior angle θ**3** and the second interior angle θ**2** may be in a range from about 0° to about 10°.

(19) The first dielectric layer **120** and the second dielectric layer **140** may be interposed between the first electrode **110**, the second electrode **130**, and the third electrode **150**. The first dielectric layer **120** and the second dielectric layer **140** may be formed of a dielectric material including silicon oxide, silicon nitride, silicon oxynitride, a high-K dielectric, or a combination thereof. The first dielectric layer **120** and the second dielectric layer **140** may both include the same dielectric material.

(20) The first dielectric layer **120** may be interposed between the first electrode **110** and the second electrode **130**. The first dielectric layer **120** may directly cover the upper surface **110**U and the side surface **110**S of the first electrode **110** in the first region A**1**, may extend to the second region A**2**, and may be in direct contact with the upper surface **10**U of the substrate **10**. The first dielectric layer **120** may cover the upper surface **110**U and the side surface **110**S of the first electrode **110** with a uniform thickness. Therefore, since the first dielectric layer **120** may allow the first electrode **110** and the second electrode **130** to be spaced apart from each other by a predetermined distance, the surface shape of the first electrode **110** may be transferred to the second electrode **130** stacked on the first electrode **110**, such that the bent region of the second electrode **130** bent in a staircase shape may be formed in a region in contact with the side surface **110**S of the first electrode **110**. This is shown by the contour of the second electrode **130** to the left of the label for the second electrode **130** in FIG. **1**, around where the first interior angle θ**1** and the second interior angle θ**2** are shown.

(21) The second dielectric layer **140** may be interposed between the second electrode **130** and the third electrode **150**. The second dielectric layer **140** may directly cover the upper surface **130**U and the side surface **130**S of the second electrode **130** in the first region A**1** and the second region A**2**, may extend to the third region A**3**, and may be in direct contact with the first dielectric layer **120**. The second dielectric layer **140** may be disposed to directly cover the upper surface **130**U and the side surface **130**S of the second electrode **130** with a uniform thickness. Accordingly, since the second dielectric layer **140** may allow the second electrode **130** and the third electrode **150** to be spaced apart from each other by a predetermined distance, the surface shape of the second electrode **130** may be transferred to the third electrode **150** stacked on the second electrode **130**, such that the bent region of the third electrode **130** bent in a staircase shape may be formed in a region in contact

with the side surface **130**S of the second electrode **130**. This is shown by the contour of the third electrode **150** to the right of the first via electrode **21** in FIG. **1**.

(22) In the metal-insulator-metal capacitor **100** in an example embodiment, the first electrode **110**, the second electrode **130**, and the third electrode **150** may be stacked in order on the substrate **10**. An upper electrode may cover a lower electrode, and damages to the lower electrode and/or the lower dielectric layer caused by misalignment of electrode patterns may be prevented in a process of etching and patterning the upper electrode. Also, since a dielectric layer interposed between the upper electrode and the lower electrode covers the side surface of the lower electrode, the process may be shortened as compared to the example in which a spacer is disposed on the side surface of the lower electrode and the dielectric layer is configured to cover the surface of the spacer.

(23) A metal-insulator-metal capacitor **200** will be described according to an example embodiment with reference to FIG. **3**. FIG. **3** is a cross-sectional diagram illustrating a metal-insulator-metal capacitor **200** according to an example embodiment. In FIG. **3** the metal-insulator-metal capacitor **200** is included in a semiconductor device **2**. In the semiconductor device **2**, the metal-insulator-metal capacitor **200** may be disposed on a substrate **10**, a cover layer **260** is disposed to cover the metal-insulator-metal capacitor **200**, and the first electrode **210**, the second electrode **230**, and the third electrode **250** of the metal-insulator-metal capacitor **200** may be electrically connected to each other by first via electrode **21** and the second via electrode **22**. The metal-insulator-metal capacitor **200** is again disposed on the substrate **10**, and also includes the first dielectric layer **220** and the second dielectric layer **240**.

(24) In the metal-insulator-metal capacitor **200** in an example embodiment, each side surface **210**S, **230**S, and **250**S of the first electrode **210**, the second electrode **230**, and the third electrode **250** may have an inclined surface, differently from the metal-insulator-metal capacitor **100** described in the aforementioned embodiment. The metal-insulator-metal capacitor **200** may be the same as the metal-insulator-metal capacitor **100** in FIG. **1** and FIG. **2** other than the above configuration. Thus, overlapping descriptions described with reference to FIG. **1** and FIG. **2** will not be provided.

(25) FIG. **3** illustrates a semiconductor device **1** in which a metal-insulator-metal capacitor **200** is employed according to an example embodiment.

(26) In the metal-insulator-metal capacitor **200** in an example embodiment, a first interior angle θ**1**′ at which the side surface **210**S and the upper surface **210**U of the first electrode **210** meet each other may be formed as an obtuse angle greater than 90°, such that the side surface **210**S may have an inclined surface. Since the first interior angle θ**1**′ of the first electrode **210** is formed as an obtuse angle, the second interior angle θ**2**′ of the second electrode **230** stacked on the first electrode **210** may be formed as an obtuse angle to correspond to the first interior angle θ**1**′ of the first electrode **210**. Also, since the second interior angle θ**2**′ of the second electrode **230** is formed as an obtuse angle, a third interior angle θ**3**′ of the third electrode **250** stacked on the second electrode **230** may also be formed as an obtuse angle to correspond to the second interior angle θ**2**′ of the second electrode **230**.

(27) A metal-insulator-metal capacitor **300** will be described according an example embodiment. FIG. **4** is a cross-sectional diagram illustrating a metal-insulator-metal capacitor **300** according to an example embodiment. In FIG. **4**, the metal-insulator-metal capacitor **300** is included in a semiconductor device **3**. In the semiconductor device **3**, the metal-insulator-metal capacitor **300** may be disposed on a substrate **10**. A cover layer **360** is disposed to cover the metal-insulator-metal capacitor **300**. The first electrode **310**, the second electrode **330**, and the third electrode **350** of the metal-insulator-metal capacitor **300** may be electrically connected to each other by the first via electrode **21** and the second via electrode **22**.

(28) In the metal-insulator-metal capacitor **300** in an example embodiment, a third dielectric layer **370** and a fourth electrode **380** may be further disposed on the first electrode **310**, the second electrode **330**, and the third electrode **350**, differently from the aforementioned example embodiments. The metal-insulator-metal capacitor **300** may have the same configuration as or a

similar configuration to that of the metal-insulator-metal capacitor **100** in FIG. **1** and FIG. **2**. Thus, overlapping descriptions described with reference to FIG. **1** and FIG. **2** will not be provided.

(29) The fourth electrode **380** may be disposed to entirely cover the upper surface **350**U and the side surface **350**S of the third electrode **350**. The fourth electrode **380** may be stacked on the third electrode **350** and may have a bent region bent in a staircase shape in a region in contact with the side surface **350**S of the third electrode **350**. The bent region of the fourth electrode **380** is shown to the right of the first via electrode **21** in FIG. **4**, and is shown with a contour to have three bent portions, or the same number of bent portions as the number of electrodes to which the fourth electrode **380** is conformed. The fourth electrode **380** may be formed of the same conductive material as that of the first electrode **310**, the second electrode **330**, and the third electrode **350**, but may be formed of a different conductive material in example embodiments. The conductive material may include a metal such as copper (Cu) and tungsten (W), a metal silicide, a metal nitride, a metal oxide, polysilicon, conductive carbon, or a combination thereof. A third groove portion **381** having a width greater than that of the second via electrode **22** may be formed in a region of the fourth electrodes **380** penetrated by the second via electrode **22**, such that the second via electrode **22** and the fourth electrode **380** may not be connected to each other. Accordingly, the first via electrode **21** may be electrically connected to the first electrode **310** and the third electrode **350**, and the second via electrode **22** may be connected to the second electrode **330** and the fourth electrode **380**.

(30) The third dielectric layer **370** may include silicon oxide, silicon nitride, silicon oxynitride, a high-K dielectric, or a combination thereof, similarly to the first dielectric layer **320** and the second dielectric layer **340**. The third dielectric layer **370** may be interposed between the third electrode **350** and the fourth electrode **380**.

(31) As described above with respect to embodiments shown in FIG. **1**, FIG. **2**, FIG. **3** and FIG. **4**, a metal-insulator-metal capacitor may include first to nth electrodes stacked on a substrate. The first electrode may be any of the first electrode **110**, the first electrode **210**, or the first electrode **310** described in these embodiments, and the nth electrode may be any of the third electrode **150**, the third electrode **250** or the fourth electrode **380** described in these embodiments. An n−1 electrode may be any of the electrodes with the second highest label number in respective embodiments, such as the third electrode **350** in the embodiment of FIG. **4** or the second electrode **130** or the second electrode **230** in other embodiments.

(32) In terms of electrodes labelled 1st to nth electrodes, the nth electrode may cover an upper surface and a side surface of the n−1 electrode stacked therebelow. First to n−1th dielectric layers are interposed between the first to nth electrodes, and cover upper surfaces and side surfaces of the first to n−1th electrodes, respectively. A first dielectric layer such as the first dielectric layer **320** in the embodiment of FIG. **4** is in contact with the substrate. Second to n−1th dielectric layers have regions in contact with first to n−2th dielectric layers stacked therebelow, respectively. An example of the n−1th dielectric layer is the third dielectric layer **370** in the embodiment of FIG. **4**.

(33) In characterizing the teachings of these embodiments using labels of 1st to nth electrodes, a relationship can be seen wherein the nth electrode has first to n−1th bent portions formed in a staircase shape in an edge region thereof. This is best seen in FIG. **4** to the right of the first via electrode **21**, wherein the contour of the fourth electrode **380** has three bent portions in a staircase shape in an edge region, though a similar relationship can be seen in FIG. **1** and FIG. **3**. The contour of the third electrode **350** in this edge region in FIG. **4** has two bent portions in a staircase shape. The contour of the second electrode **330** in this edge region in FIG. **4** has one bent portion in a staircase shape. An interior angle between the upper surface and the side surface of the nth electrode may be in a range of 90° to 135°. For example, the interior angles shown in FIG. **1** and FIG. **4** are approximately 90°, whereas the interior angles shown in FIG. **3** are approximately 135°. An interior angle between the upper surface and the side surface of the nth electrode and an interior angle between the upper surface and the side surface of the n−1th electrode may have an angle

difference of 0° to 10°, which is to say that the electrodes are fitted to substantially conform to the electrodes immediately below.

(34) Moreover, the nth electrode may have a size larger than a size of the n−1th electrode. Each of the first to n−1th dielectric layers may have a substantially uniform thickness. For the nth electrode with the first to n−1th bent portions formed in a staircase shape in an edge region, the first to n−1th bent portions may have a substantially equal difference therebetween, such as by having substantially equal interior angles.

(35) Moreover, the nth electrode may have at least one groove portion in which the n−1th dielectric layer is disposed on a bottom surface thereof in a region overlapping the n−1th electrode. Examples of groove portions include the first groove portion **131**, the second groove portion **151**, and the third groove portion **381**.

(36) Accordingly, whether the nth electrode in the above descriptions is seen as the fourth electrode **380** in FIG. **4** or the third electrode **150** in FIG. **1** and/or the third electrode **250** in FIG. **3**, the characteristics of some of the relative sizes and angle differences may be seen as scalable in embodiments with different numbers of total electrodes and dielectric layers.

(37) A method of manufacturing a package substrate will be described with reference to FIG. **5**, FIG. **6**, FIG. **7**, FIG. **8**, FIG. **9**, FIG. **10**, FIG. **11**, and FIG. **12**. FIG. **5**, FIG. **6**, FIG. **7**, FIG. **8**, FIG. **9**, FIG. **10**, FIG. **11** and FIG. **12** are each cross-sectional diagrams illustrating a method of manufacturing the metal-insulator-metal capacitor **100** illustrated in FIG. **1**.

(38) Referring to FIG. **5**, a first electrode **110** may be formed on an upper surface **10**U of a substrate **10**. The first electrode **110** may be formed by depositing a conductive material on the upper surface **10**U of the substrate **10**. The conductive material may include a metal such as aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt) and gold (Au), metal silicide, metal nitride, metal oxide, polysilicon, conductive carbon, or a combination thereof. The side surface **110**S of the first electrode **110** may be formed to have a first interior angle θ**1** of about 90° to about 135° with the upper surface **110**U.

(39) Referring to FIG. **6**, the first dielectric layer **120** may be formed to cover the first electrode **110**. The first dielectric layer **120** may be formed to cover the upper surface **110**U and the side surface **110**S of the first electrode **110**. The first dielectric layer **120** may include silicon oxide, silicon nitride, silicon oxynitride, a high-K dielectric, or a combination thereof. The first dielectric layer **120** may be conformally formed on the first electrode **110** by a predetermined thickness. Accordingly, in a subsequent process of forming the second electrode **130**, an exterior shape of the first electrode **110** may be transferred to the second electrode **130**.

(40) Referring to FIG. **7**, the second electrode **130** may be formed on the surface of the first dielectric layer **120**. The second electrode **130** may be formed to have a size larger than that of the first electrode **110** and may entirely cover the first electrode **110**. At least one instance of a first groove portion **131** exposing the first dielectric layer **120** may be formed on a bottom surface of the second electrode **130** by etching a region of the second electrode **130** overlapping the first electrode **110**. The second electrode **130** may be formed by depositing the same conductive material as that of the first electrode **110**, but may be formed by depositing a conductive material different than that of the first electrode **110** in example embodiments.

(41) Referring to FIG. **8**, the second dielectric layer **140** may be formed to cover the second electrode **130**. The second dielectric layer **140** may be formed to cover an upper surface and a side surface of the second electrode **130**. Accordingly, the second dielectric layer **140** may also be formed on the side surface and the bottom surface of the first groove portion **131**. Similarly to the first dielectric layer **120**, the second dielectric layer **140** may include silicon oxide, silicon nitride, silicon oxynitride, a high-K dielectric, or a combination thereof. The second dielectric layer **140** may be conformally formed on the second electrode **130** by a predetermined thickness. Accordingly, in a subsequent process of forming the third electrode **150**, an exterior shape of the second electrode **130** may be transferred to the third electrode **150**.

(42) Referring to FIG. **9**, the third electrode **150** may be formed to cover the surface of the second dielectric layer **140**. The third electrode **150** may be formed to have a size larger than that of the second electrode **130** and may entirely cover the second electrode **130**.
(43) At least one instance of a one second groove portion **151** through which the second dielectric layer **140** is exposed may be formed on a bottom surface of the third electrode **150** by etching a region of the third electrode **150** overlapping the second electrode **130**. The third electrode **150** may be formed by depositing the same conductive material as that of the first electrode **110** and the second electrode **130**, but may be formed by depositing a conductive material different than that of the first electrode **110** and the second electrode **130** in example embodiments.
(44) Referring to FIG. **10**, a cover layer **160** may be formed to entirely cover the third electrode **150**. The cover layer **160** may be formed by depositing a dielectric material, similarly to the first dielectric layer **120** and the second dielectric layer **140**. The dielectric material may include silicon oxide, silicon nitride, silicon oxynitride, a high-k dielectric, or a combination thereof. The cover layer **160** may be planarized by performing a chemical mechanical polishing (CMP) process on the surface after applying a dielectric material to cover the third electrode **150**.
(45) Referring to FIG. **11**, a via hole V extending inwardly may be formed. The via hole V may be formed to extend from the surface of the cover layer **160** into the substrate **10**.
(46) Referring to FIG. **12**, the first via electrode **21** and the second via electrode **22** may be formed by filling the via hole V. The first via electrode **21** and the second via electrode **22** may be formed in a columnar shape filling the via hole V. The first via electrode **21** may be electrically connected to the first electrode **110** and the third electrode **150**, and the second via electrode **22** may be electrically connected to the second electrode **130**. Thereafter, a passivation layer **30** and a protective layer **40** may be formed to cover the first via electrode **21** and the second via electrode **22** and the cover layer **160**, and a hole **50** penetrating through the passivation layer **30** and the protective layer **40** may be formed (see FIG. **1**).
(47) According to the aforementioned example embodiments, the upper electrode may cover the upper surface and the side surface of the lower electrode, such that damages to the lower electrode may be prevented in the manufacturing process, and a metal-insulator-metal capacitor having improved reliability and yield may be provided.
(48) While the example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A metal-insulator-metal capacitor, comprising: a first electrode in contact with an upper surface of a substrate and disposed in a first region of the upper surface; a second electrode covering the first electrode and extending to a second region, the second region surrounding an outer periphery of the first region; a third electrode covering the second electrode and extending to a third region, the third region surrounding an outer periphery of the second region; a first dielectric layer disposed between the first electrode and the second electrode to cover an upper surface and a side surface of the first electrode and extending to the second region of the upper surface of the substrate; and a second dielectric layer disposed between the second electrode and the third electrode to cover an upper surface and a side surface of the second electrode and extending to the third region of the upper surface of the substrate and in contact with the first dielectric layer; wherein the first dielectric layer contacts an upper surface of the substrate and extends parallel to the upper surface of the substrate in the third region, wherein the second dielectric layer contacts an upper surface of the first dielectric layer and extends parallel to an upper surface of the first dielectric layer in the third region, and wherein the third electrode contacts an upper surface of the second dielectric layer in the third region.

2. The metal-insulator-metal capacitor of claim 1, wherein the first dielectric layer covers the upper surface and the side surface of the first electrode with a substantially uniform thickness, and wherein the second dielectric layer covers the upper surface and the side surface of the second electrode with a substantially uniform thickness.
3. The metal-insulator-metal capacitor of claim 1, wherein the second electrode includes at least one first groove portion disposed in the first region and penetrating through the second electrode.
4. The metal-insulator-metal capacitor of claim 3, wherein the third electrode includes at least one second groove portion disposed in the second region and penetrating through the third electrode.
5. The metal-insulator-metal capacitor of claim 4, further comprising: a first via electrode spaced apart from an internal side wall of the first groove portion, penetrating through the first groove portion, and in contact with the first electrode and the third electrode; and a second via electrode spaced apart from an internal side wall of the second groove portion, penetrating through the second groove portion, and in contact with the second electrode.
6. The metal-insulator-metal capacitor of claim 1, wherein an interior angle between the upper surface and the side surface of the first electrode is in a range of 90° to 135°.
7. The metal-insulator-metal capacitor of claim 6, wherein an interior angle between the upper surface and the side surface of the second electrode and an interior angle between the upper surface and the side surface of the first electrode have an angle difference of 0° to 10°.
8. The metal-insulator-metal capacitor of claim 1, wherein the first electrode, the second electrode and the third electrode include substantially the same conductive material, and wherein the same conductive material included in the first electrode, the second electrode and the third electrode includes at least one of aluminum (Al), copper (Cu), nickel (Ni), tungsten (W), platinum (Pt), and gold (Au).
9. The metal-insulator-metal capacitor of claim 1, wherein the first dielectric layer and the second dielectric layer both include the same dielectric material.
10. A metal-insulator-metal capacitor, comprising: a first electrode disposed on an upper surface of a substrate and disposed in a first region of the upper surface; a second electrode covering an upper surface and a side surface of the first electrode and extending to a second region, the second region surrounding the first region; a third electrode covering a side surface and an upper surface of the second electrode and extending to a third region, the third region surrounding the second region; a first dielectric layer interposed between the first electrode and the second electrode; and a second dielectric layer interposed between the second electrode and the third electrode, wherein the first dielectric layer contacts an upper surface of the substrate and extends parallel to the upper surface of the substrate in the third region, wherein the second dielectric layer contacts an upper surface of the first dielectric layer and extends parallel to an upper surface of the first dielectric layer in the third region, wherein the third electrode contacts an upper surface of the second dielectric layer in the third region, and wherein an interior angle between the upper surface and the side surface of each of the first electrode, the second electrode and the third electrode is in a range of 90° to 135°.
11. The metal-insulator-metal capacitor of claim 10, wherein an interior angle between the upper surface and the side surface of the second electrode and an interior angle between the upper surface and the side surface of the first electrode have an angle difference of 0° to 10°.
12. The metal-insulator-metal capacitor of claim 10, wherein the third electrode has a size larger than a size of the second electrode, and wherein the second electrode has a size larger than a size of the first electrode.
13. The metal-insulator-metal capacitor of claim 10, further comprising: a cover layer covering the third electrode.
14. The metal-insulator-metal capacitor of claim 1, wherein the second electrode extends to the second region without extending to the third region.
15. The metal-insulator-metal capacitor of claim 1, wherein only the third electrode among the first electrode, the second electrode, and the third electrode, is disposed in the third region.